

# JEE MAIN REVISION MASTER: ALTERNATING CURRENT 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                | Frequency | Category               |
|---------------|---------------------------------------------|-----------|------------------------|
| Very High     | Series LCR Circuit & Impedance              | 16 PYQs   | Circuits               |
| High          | AC Fundamentals: Peak, RMS & Phase          | 13 PYQs   | Foundations            |
| High          | Reactance: Inductive & Capacitive           | 11 PYQs   | Impedance Elements     |
| Moderate      | Resonance, Q-Factor & Bandwidth             | 6 PYQs    | Resonance              |
| Moderate      | Power in AC Circuits & Power Factor         | 4 PYQs    | Power & Efficiency     |
| Moderate      | LC Oscillations & Transformers              | 3 PYQs    | Devices & Oscillations |
| Moderate      | Kcet Alternating Current - Core Application | 3 PYQs    | General Concept        |

## 2. Master Formula Reference Card

| Concept / Topic                     | Mathematical Formulation & Governing Equations                                                                                                                                                      |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Series LCR Circuit & Impedance      | $Z = \sqrt{R^2 + (X_L - X_C)^2}$ , $\tan\phi = \frac{X_L - X_C}{R}$ , $V_{\text{total}} = \sqrt{V_R^2 + (V_L - V_C)^2}$                                                                             |
| AC Fundamentals: Peak, RMS & Phase  | $I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$ , $V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$ , $I_{\text{avg, half}} = \frac{2 I_0}{\pi}$                                                                      |
| Reactance: Inductive & Capacitive   | $X_L = \omega L = 2\pi f L$ , $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$                                                                                                                       |
| Resonance, Q-Factor & Bandwidth     | $\omega_0 = \frac{1}{\sqrt{LC}}$ , $Z_{\min} = R$ , $I_{\max} = \frac{V}{R}$ , $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{\omega_0}{\Delta\omega}$ |
| Power in AC Circuits & Power Factor | $P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi$ , $\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$ , $I_{\text{wattless}} = I_{\text{rms}} \sin\phi$                       |
| LC Oscillations & Transformers      | $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$ , $\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{V_s I_s}{V_p I_p}$ , $q(t) = q_0 \cos(\omega_0 t)$                                     |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Series LCR Circuit & Impedance (Very High Yield — 16 PYQs)

**Core Principle:** Series LCR impedance  $Z = \sqrt{R^2 + (X_L - X_C)^2}$ . Circuit current  $I = (V / Z)$ . Phase angle  $\tan\phi = (X_L - X_C / R)$ . Net voltage phasor  $V_{\text{(total)}} = \sqrt{V_R^2 + (V_L - V_C)^2}$ .

■ **Caution / JEE Pitfall:** Voltages across series AC elements do NOT add algebraically:  $V_{\text{(total)}} \neq V_R + V_L + V_C$ ! Always use phasor vector addition:  $V_{\text{(total)}} = \sqrt{V_R^2 + (V_L - V_C)^2}$ .

■ **Speed Trick:** If  $X_L > X_C$ , circuit is inductive (voltage leads current by  $\phi$ ). If  $X_C > X_L$ , circuit is capacitive (current leads voltage by  $\phi$ ).

**Representative Source PYQs:**

- [2026 Shift] Q#3 (MCQ): "In a circuit containing a pure resistor connected to an AC source,..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "A series LCR circuit containing an AC source of 100 V has an inductor and a capacitor of reactances  $24\Omega$  and  $16\Omega$  respectively. ..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "The figure shows variation of  $R, X_L$  and  $X_C$  with frequency  $f$  in a series L-C-R circuit. Then, for what frequency point is the ci..." [Ans: N/A]

### 3.2 AC Fundamentals: Peak, RMS & Phase (High Yield — 13 PYQs)

**Core Principle:** Sinusoidal AC quantities:  $I_{\text{(rms)}} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0$ ,  $V_{\text{(rms)}} = \frac{V_0}{\sqrt{2}}$ . Average current over full cycle is 0; average over half cycle is  $I_{\text{(avg, half)}} = (2I_0 / \pi) \approx 0.637 I_0$ .

■ **Caution / JEE Pitfall:** AC voltmeters and ammeters measure RMS values, not peak values! When JEE specifies '220 V AC', it always denotes  $V_{\text{(rms)}}$  (peak voltage is  $V_0 = 220\sqrt{2} \approx 311 \text{ V}$ ).

■ **Speed Trick:** Instantaneous power  $p(t) = v(t)i(t)$  fluctuates at double frequency ( $2\omega$ ) about average power  $P_{\text{(avg)}} = V_{\text{(rms)}}I_{\text{(rms)}}\cos\phi$ .

**Representative Source PYQs:**

- [2026 Shift] Q#2 (MCQ): "A light bulb rated 100 W is connected to an AC source of 220 V, 50 Hz. The rms current through the bulb is..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "In a circuit containing a pure resistor connected to an AC source,..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "A sinusoidal voltage produced by an AC generator at any instant  $t$  is given by an equation  $V = 311\sin 314 t$ . The rms value of voltage and freq..." [Ans: N/A]

### 3.3 Reactance: Inductive & Capacitive (High Yield — 11 PYQs)

**Core Principle:** Pure Inductor: Voltage leads current by  $90^\circ$  ( $(\pi / 2)$ ), inductive reactance  $X_L = \omega L = 2\pi f L$ . Pure Capacitor: Current leads voltage by  $90^\circ$ , capacitive reactance  $X_C = (1 / \omega C) = (1 / 2\pi f C)$ .

■ **Caution / JEE Pitfall:** In direct current (DC,  $f = 0$ ), an inductor offers zero reactance ( $X_L = 0$ , short circuit) while a capacitor offers infinite reactance ( $X_C \rightarrow \infty$ , blocking DC completely).

■ **Speed Trick:** Mnemonic 'CIVIL': In a Capacitor (C), Current (I) leads Voltage (V); in an Inductor (L), Voltage (V) leads Current (I).

**Representative Source PYQs:**

- [2025 Shift] Q#2 (MCQ): "A series LCR circuit containing an AC source of 100 V has an inductor and a capacitor of reactances  $24\Omega$  and  $16\Omega$  respectively. ..." [Ans: N/A]
- [2022 Shift] Q#1 (MCQ): "In a series LCR circuit,  $R = 300\Omega, L = 0.9H, C = 2.0\mu F$  and  $\omega = 1000 \text{ rad/s}$ , then impedance of the circuit is..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "A series resonant AC circuit contains a capacitance  $10^{-6} \text{ F}$  and an inductor of  $10^{-4} \text{ H}$ . The frequency of electrical oscillations wi..." [Ans: N/A]

### 3.4 Resonance, Q-Factor & Bandwidth (Moderate Yield — 6 PYQs)

**Core Principle:** At resonance:  $X_L = X_C$ , resonant frequency  $\omega_0 = \frac{1}{\sqrt{LC}}$  ( $f_0 = \frac{1}{2\pi\sqrt{LC}}$ ). Minimum impedance  $Z_{\min} = R$ , maximum current  $I_{\max} = (V / R)$ . Quality factor  $Q = (\omega_0 L / R) = (1 / \omega_0 C R) = (1 / R) \sqrt{(L / C)} = (\omega_0 / \Delta\omega)$ .

■■ **Caution / JEE Pitfall:** At resonance,  $V_L$  and  $V_C$  are NOT zero! In fact,  $V_L = V_C = Q \cdot V_{\text{supply}}$ , which can be substantially higher than supply voltage (voltage magnification)!

■ **Speed Trick:** Resonance bandwidth  $\Delta\omega = (R / L)$ . To sharpen resonance (higher  $Q$ ), decrease resistance  $R$  or increase the  $(L / C)$  ratio.

**Representative Source PYQs:**

- **[2024 Shift] Q#3 (MCQ):** "Which of the following combinations should be selected for better tuning of an L-C-R circuit used for communication?..." **[Ans: N/A]**
- **[2024 Shift] Q#4 (MCQ):** "In an L-C-R series circuit, the value of only capacitance  $C$  is varied. The resulting variation of resonance frequency  $f_0$  as a funct..." **[Ans: C]**
- **[2023 Shift] Q#2 (MCQ):** "Total impedance of a series L - C - R circuit varies with angular frequency of the AC source connected to it as shown in the graph. ..." **[Ans: N/A]**

### 3.5 Power in AC Circuits & Power Factor (Moderate Yield — 4 PYQs)

**Core Principle:** Average power dissipated  $P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi$ . Power factor  $\cos\phi = (R / Z) = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$ . Wattless current component  $I_{\text{wattless}} = I_{\text{rms}} \sin\phi$ .

■■ **Caution / JEE Pitfall:** Power factor is  $\cos\phi = (R / Z)$ , NOT  $\tan\phi$  or  $(R / X)$ ! For pure L or pure C,  $\phi = 90^\circ$  implies  $\cos\phi = 0$ , so average power dissipation is strictly zero.

■ **Speed Trick:** Apparent power  $S = V_{\text{rms}} I_{\text{rms}}$  (in VA). True power  $P = S \cos\phi$  (in W). Reactive power  $Q = S \sin\phi$  (in VAR).

**Representative Source PYQs:**

- **[2020 Shift] Q#1 (MCQ):** "The power factor of R-L circuit is  $\frac{1}{\sqrt{3}}$ . If the inductive reactance is  $2\Omega$ . The value of resistance is..." **[Ans: N/A]**
- **[2019 Shift] Q#1 (MCQ):** "An inductor of inductance  $L$  and resistor  $R$  are joined together in series and connected by a source of frequency  $\omega$ . The power dis..." **[Ans: N/A]**
- **[2018 Shift] Q#3 (MCQ):** "A 100 W bulb is connected to an AC source of 220V, 50Hz. Then, the current flowing through the bulb is..." **[Ans: N/A]**

### 3.6 LC Oscillations & Transformers (Moderate Yield — 3 PYQs)

**Core Principle:** Ideal transformer equations:  $(V_s / V_p) = (N_s / N_p) = (I_p / I_s)$ . Efficiency  $\eta = \frac{P_{out}}{P_{in}} = (V_s I_s / V_p I_p)$ . LC tank oscillations: charge  $q(t) = q_0 \cos(\omega_0 t)$  with total energy conserved  $U = (q^2 / 2C) + (1/2) L I^2$ .

■■ **Caution / JEE Pitfall:** Transformers operate exclusively on AC; in DC, primary flux is static ( $(d\Phi / dt) = 0$ ), producing zero secondary voltage and potentially burning the primary coil!

■ **Speed Trick:** In step-up transformers ( $N_s > N_p$ ), voltage increases but current decreases, keeping total power  $V I$  conserved under ideal conditions.

**Representative Source PYQs:**

- **[2026 Shift] Q#1 (MCQ):** "A small town with a demand of 900kW of electric power at 220V is situated 20km away from an electric power genera..." [Ans: N/A]
- **[2023 Shift] Q#4 (MCQ):** "An ideal transformer has a turns ratio of 10. When the primary is connected to 220V,50Hz as source, the power output is..." [Ans: N/A]
- **[2017 Shift] Q#4 (MCQ):** "The output of a step down transformers is measured to be 48 V when connected to a 12 W bulb. The value of peak current is..." [Ans: N/A]

### 3.7 Kcet Alternating Current - Core Application (Moderate Yield — 3 PYQs)

**Core Principle:** Key problem pattern in Kcet Alternating Current.

■■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

**Representative Source PYQs:**

- **[2025 Shift] Q#4 (MCQ):** "In domestic electric mains supply, the voltage and the current are..." [Ans: N/A]
- **[2021 Shift] Q#2 (MCQ):** "LC-oscillations are similar and analogous to the mechanical oscillations of a block attached to a spring. The electrical equivalent of the f..." [Ans: N/A]
- **[2017 Shift] Q#3 (MCQ):** "In the A.C. circuit shown, keeping ' K ' pressed, if an iron rod inserted into the coil, the bulb in the circuit,..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

**Benchmark Problem 1: KCET 2026 (Q#1)**

A small town with a demand of 900kW of electric power at 220V is situated 20km away from an electric power generating station. Each wire of the two-wire transmission line has a resistance per unit length of  $5 \times 10^{-4} \Omega \text{m}^{-1}$ . The town gets power from the line through a 45000V to 220V step-down transformer at a substation in the town. The line power loss in the form of heat is:

- (A) 4 kW
- (B) 8 kW
- (C) 40 kW
- (D) 80 kW

**Correct Answer: None**

**Solution Analysis:** The town receives power through a step-down transformer whose primary voltage is 45000V. The power demand of the town is  $P=900\text{kW}=900000\text{W}$ . So the current in the transmission line is  $I=(P/V)$   $I=(900000/45000)=20\text{A}$ . Now, each wire is 20km long.  $20\text{km}=20000\text{m}$ . Resistance per unit length of each wire is  $5 \times 10^{-4} \Omega \text{m}^{-1}$ . So resistance of one wire is  $R=5 \times 10^{-4} \times 20000$   $R=10\Omega$ . Since it is a two-wire transmission line, total resistance is  $R_{\text{total}}=10+10=20\Omega$ . Power loss in the line

**Benchmark Problem 2: KCET 2026 (Q#2)**

A light bulb rated 100 W is connected to an AC source of 220 V, 50 Hz. The rms current through the bulb is

- (A) 0.454 A
- (B) 0.545 A
- (C) 2.20 A
- (D) 0.22 A

**Correct Answer: None**

**Solution Analysis:** For a light bulb, the given power rating is  $P=100\text{W}$  and the AC voltage given is the rms voltage,  $V_{\text{rms}}=220\text{V}$ . For a bulb, which behaves like a resistor, the power is  $P=V_{\text{rms}}I_{\text{rms}}$ . So,  $I_{\text{rms}}=\frac{P}{V_{\text{rms}}}$ . Substitute the values:  $I_{\text{rms}}=(100/220)$   $I_{\text{rms}}=0.454\text{A}$ . So, the rms current through the bulb is 0.454A. Correct option: Option A.

**Benchmark Problem 3: KCET 2025 (Q#2)**

A series LCR circuit containing an AC source of 100 V has an inductor and a capacitor of reactances  $24\Omega$  and  $16\Omega$  respectively. If a resistance of  $6\Omega$  is connected in series, then the potential difference across the series combination of inductor and capacitor only is

- (A) 80 V
- (B) 400 V
- (C) 8 V
- (D) 40 V

**Correct Answer: None**

**Solution Analysis:**  $X_C=24\Omega$   $X_L=16\Omega$   $R=6\Omega$   $Z=\sqrt{R^2+(X_C-X_L)^2}=\sqrt{6^2+8^2}=\sqrt{100}=10\Omega$   $i=(V/Z)=(100/10)=10\text{A}$ . Net voltage across capacitor and inductor  $=i(X_C-X_L)=10(24-16)=10(8)=80\text{V}$

**Benchmark Problem 4: KCET 2024 (Q#1)**

The figure shows variation of  $R, X_L$  and  $X_C$  with frequency  $f$  in a series L-C-R circuit. Then, for what frequency point is the circuit capacitive?

- (A)
- (B)
- (C)
- (D)

**Correct Answer: None**

**Solution Analysis:** For a capacitive circuit,  $X_C > X_L$ . For the given diagram in question, the value  $X_C$  corresponding to point A is greater than the value of  $X_L$ .

**Benchmark Problem 5: KCET 2021 (Q#2)**

LC-oscillations are similar and analogous to the mechanical oscillations of a block attached to a spring. The electrical equivalent of the force constant of the spring is

- (A) reciprocal of capacitive reactance
- (B) capacitive reactance
- (C) reciprocal of capacitance
- (D) capacitance

**Correct Answer: None**

**Solution Analysis:** For LC-oscillations,  $L \frac{di}{dt} + \frac{q}{C} = 0$ . Differentiating further it w.r.t  $t$ , we get  $L \frac{d^2i}{dt^2} + \frac{1}{C} i = 0$ .... (i) Similarly, for spring-block system,  $m \frac{d^2x}{dt^2} + kx = 0$ ..... (ii) Comparing Eqs. (i) and (ii), we get  $m \propto L$  and  $k \propto (1/C)$